

EXERCISE

1. Show that ${}_1F_1(\beta; \gamma; x) = \lim_{\beta \rightarrow \infty} {}_2F_1(\alpha, \beta; \gamma; x/\beta)$.

2. Show that ${}_2F_1(\alpha, \beta; \beta - \alpha + 1; -1) = \frac{\Gamma(1 + \beta - \alpha)\Gamma(1 + \beta/2)}{\Gamma(1 + \beta)\Gamma(1 + \beta/2 - \alpha)}$ and deduce that

$${}_2F_1(\alpha, 1 - \alpha; \gamma; 1/2) = \frac{\Gamma(\gamma/2)\Gamma(\gamma/2 + 1/2)}{\Gamma(\alpha/2 + \gamma/2)\Gamma(1/2 - \alpha/2 + \gamma/2)}.$$

3. Evaluate the integral $\int_0^{\infty} e^{-sx} {}_1F_1(\alpha; \beta; x) dx$. [Ans. $(1/s)_2 \times F_1(\alpha, 1; \beta; s)$]

[Hint. Use Art. 14.15]

4. Prove that $F(\alpha, \beta + 1; \gamma + 1; x) - F(\alpha, \beta; \gamma; x) = \frac{\alpha(\gamma - \beta)}{\gamma(\gamma + 1)} x F(\alpha + 1, \beta + 1; \gamma + 2; x)$.

5. The complete elliptic integral of the first kind is

$$K = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}. \text{ Show that } K = \frac{\pi}{2} F\left(\frac{1}{2}, \frac{1}{2}; 1; k^2\right).$$

6. The complete elliptic integral of the second kind is

$$E = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \phi} d\phi. \text{ Show that } E = \frac{\pi}{2} F\left(-\frac{1}{2}, \frac{1}{2}; 1; k^2\right), |k| < 1.$$

7. Prove the following relations :

$$(i) F(\alpha - 1, \beta - 1; \gamma; x) - F(\alpha, \beta - 1; \gamma; x) = \frac{(1 - \beta)x}{\gamma} F(\alpha, \beta; \gamma + 1; x)$$

$$(ii) \alpha F(\alpha + 1, \beta; \gamma; x) - (\gamma - 1)F(\alpha, \beta; \gamma - 1; x) = (\alpha + 1 - \gamma)F(\alpha, \beta; \gamma; x).$$

8. Prove that $F(\alpha, \beta; \gamma; 1/2) = 2^\alpha F(\alpha, \gamma - \beta; \gamma; -1)$.

9. Show that (i) $e^x - 1 = xF(1; 2; x)$.

$$(ii) (1 + x/\alpha)e^x = F(\alpha + 1; \alpha; x).$$

10. The incomplete Gamma function is defined by the equation $\gamma(\alpha, x) = \int_0^x e^{-t} t^{\alpha-1} dt, \alpha > 0$.

Prove that

$$\gamma(\alpha, x) = \alpha^{-1} x^\alpha F(\alpha; \alpha + 1; -x).$$

11. Prove that following relations : (i) $\beta F(\alpha; \beta; x) = \beta F(\alpha - 1; \beta; x) + xF(\alpha; \beta + 1; x)$.

$$(ii) \alpha F(\alpha + 1; \beta; x) - (\beta - 1)F(\alpha; \beta - 1; x) = (\alpha - \beta + 1)F(\alpha; \beta; x).$$

12. Prove the following relations :

$$(i) F(\alpha, \beta; \gamma; x) - F(\alpha, \beta; \gamma - 1; x) = -\frac{\alpha\beta x}{\gamma(\gamma - 1)} F(\alpha + 1, \beta + 1; \gamma + 1; x)$$

$$(ii) F(\alpha + 1; \gamma; x) - F(\alpha; \gamma; x) = (x/\gamma) \times F(\alpha + 1, \gamma + 1; x).$$

13. Hypergeometric function ${}_2F_1(\alpha, \beta; \gamma; x) = 1 + \frac{\alpha \cdot \beta}{1 \cdot \gamma} x + \frac{\alpha(\alpha + 1) \cdot \beta(\beta + 1)}{1 \cdot 2 \cdot \gamma(\gamma + 1)} x^2 + \dots$ is the

solution of the differential equation $x(1 - x)y'' + [\gamma - (\alpha + \beta + 1)x]y' - \alpha\beta y = 0$. Show that

$$(a) \left[\frac{d}{dx} {}_2F_1(\alpha, \beta; \gamma; x) \right]_{x=0} = \frac{\alpha\beta}{\gamma}. \quad (b) {}_2F_1\left(\alpha, \beta; \frac{1}{2}\alpha + \frac{1}{2}\beta + \frac{1}{2}; \frac{1}{2}\right) = \frac{\Gamma(\frac{1}{2})\Gamma(\frac{1}{2}\alpha + \frac{1}{2}\beta + \frac{1}{2})}{\Gamma(\frac{1}{2} + \frac{1}{2}\alpha)\Gamma(\frac{1}{2} + \frac{1}{2}\beta)}.$$

$$(c) {}_2F_1\left(\alpha, \beta; \gamma; \frac{1}{2}\right) = 2^\alpha {}_2F_1(\alpha, \gamma - \beta; \gamma; -1).$$

14. Prove that (a) $P_n(\cos \theta) = {}_2F_1(-n; n + 1; 1; \sin^2 \theta/2)$.

$$(b) P_n(\cos \theta) = (-1)^n {}_2F_1(n + 1, -n; 1; \cos^2 \theta/2).$$

$$(c) P_n(x) = \left(\frac{x-1}{2}\right)^n {}_2F_1\left(-n, -n; 1; \frac{x+1}{x-1}\right).$$

15. Show that

$$(a) H_{2n}(x) = (-1)^n \frac{(2n!)}{n!} {}_1F_1\left(-n; \frac{1}{2}; x^2\right). \quad (b) H_{2n+1}(x) = (-1)^n \frac{2(2n+1)!}{n!} x {}_1F_1\left(-n; \frac{3}{2}; x^2\right).$$

16. Show that $L_n(x) = n! {}_1F_1(-n; 1; x)$.

17. Prove that, for $|x| < a$, ${}_2F_1(a, b, c, x) = (1 - x)^{c-a-b} {}_2F_1(c - a, c - b, c, x)$

18. State the confluent hyper geometric equation and explain its solution.

EXERCISE

1. Show that $y = x^{-n/2} J_n(2\sqrt{x})$ satisfies the equation $xy'' + (n+1)y' + y = 0$.

2. Show that $\int_0^{\pi/2} \sqrt{(\pi x)} J_{1/2}(2x) dx = 1$.

3. (a) Prove : $\int_0^\infty J_n(bx) x^n e^{-ax} dx = \frac{2^n \Gamma(n + \frac{1}{2})}{\sqrt{\pi}} \cdot \frac{b^n}{(a^2 + b^2)^{n+1/2}}$

(b) Prove that $\int_0^\infty J_n(bx) x^{n+1} e^{-ax} dx = \frac{2^{n+1} \Gamma(n + \frac{1}{2})}{\pi} \cdot \frac{ab^n}{(a^2 + b^2)^{n+3/2}}, a > 0$.

4. Prove : $\int_0^\infty J_n(bx) x^{n+1} e^{-ax^2} dx = \frac{b^n}{(2a)^{n+1}} \exp(-b^2 / 4a)$, where $\exp(p) = e^p$

5. Prove that $\int_0^\infty J_n(bx) x^{n+1} e^{-ax^2} dx = \frac{b^n}{2^{n+1} a^{n+2}} \left(n+1 - \frac{b^2}{4a} \right) \exp\left(-\frac{b^2}{4a}\right), a > 0$.

6. For what value of n the general solution of Bessel's differential equation will be of the form $y = AJ_n(x) + BJ_n(x)$.

7 Write the differential equation satisfied by Bessel's function of order n . Express the following Bessel's functions in terms of trigonometric functions :

(i) $J_{1/2}(x)$,

(ii) $J_{-1/2}(x)$,

(iii) $J_{3/2}(x)$,

(iv) $J_{-3/2}(x)$.

EXERCISE

1. Show that
$$\sum_{n=0}^{\infty} P_{n+m}^m(x) t^n = \frac{(2m)!(1-x^2)^{m+1/2}}{2^m m!(1-2tx+t^2)^{m+1/2}}.$$
2. Prove that (a) $P_n^m(x) = \frac{1}{2^n \cdot n!} (1-x^2)^{m/2} \frac{d^{n+m}}{dx^{n+m}} (x^2-1)^n$ (b) $P_n^{-m}(x) = (-1)^m \frac{(n-m)!}{(n+m)!} P_n^m(x).$
3. Prove that
$$P_n^m(x) = \frac{(n+m)!(1-x^2)^{m/2}}{(n-m)!2^m m!} {}_2F_1\left(m-n, m+n+1; m+1; \frac{1-x}{2}\right).$$
4. Define associated Legendre's polynomials and prove their orthogonality condition.
5. Prove that
$$P_m^m(x) = \frac{(n+m)!}{(n-m)!} \times \frac{(1-x^2)^{m/2}}{2^m m!} {}_2F_1\left(m-n, m+n+1; m+1; \frac{1-x}{2}\right).$$
6. Prove that
$$P_n^m(-x) = (-1)^{m+n} P_n^m(x)$$

PROBLEMS

Ex. 1. Find the series solutions of the following equation about $x = 0$.

(i) $2x^2 y'' + xy' + (x^2 - 1)y = 0$

Ans. $y = ax(1 - x^2/14 + x^4/616 + \dots) + bx^{-1/2}(1 - x^2/2 + x^4/40 + \dots)$

(ii) $x^2 y'' - xy' + (x^2 + 8/9)y = 0$

Ans. $y = ax^{4/3}(1 - 3x^2/16 + 9x^4/896 + \dots) + bx^{2/3}(1 - 3x^2/8 + 9x^4/320 + \dots)$

(iii) $xy'' + y' - y = 0$

Ans. $y = (a + b \log x)(1 + x + x^2/(2!)^2 + \dots) - 2b\{x + (x^2/(2!)^2) \times (1 + 1/2) + \dots\}$

(iv) $y'' + x^2 y = 0$ **Ans.** $y = a\left(1 - \frac{x^4}{4 \cdot 3} + \frac{x^8}{8 \cdot 7 \cdot 4 \cdot 3} + \dots\right) + b\left(x - \frac{x^5}{5 \cdot 4} + \frac{x^9}{9 \cdot 8 \cdot 5 \cdot 4} \dots\right)$

Ex. 2. Find the power series solution of $xy'' + y' + 2y = 0$ about $x = 1$, i.e., in power of $(x - 1)$.

Ans. $y = a\{1 - (x-1)^2 + (2/3) \times (x-1)^3 - (1/3) \times (x-1)^4 + \dots\} + b\{(x-1) - (1/2) \times (x-1)^2 + (1/12) \times (x-1)^4 + \dots\}$

Ex. 3. Find the general solution of the differential equation $(x-1)^2 y'' + (x-1)y' - 4y = 0$ in powers of x using the Frobenius method.

Ex. 4. Find the general solution of $(1+x^2)y'' + 2xy' - 2y = 0$ in terms of power series in x .

Ex. 5. Show that the differential equation $x^2 y'' - 3xy' + (4x+4)y = 0$ has only one Frobenius series solution and find it.

Find the series solutions of the following equations near $x = 0$.

1. $xy'' + (1+x)y' + 2y = 0$.

Ans. $y = a(1 - 2x + (3/2!)x^2 - (4/3!)x^3 + \dots) + b[\log x \{1 - 2x + (3/2!) \times x^2 - (4/3!) \times x^3 + \dots\} + 2(2 - 1/2)x - (3/2!) \times (-1/3 + 2 + 1/2)x^2 + \dots]$

2. $x^2y'' - x(1+x)y' + y = 0$.

Ans. $y = ax[1 + x + (1/2) \times x^2 + (1/2 \cdot 3) \times x^3 + \dots] + b[\log x \{1 + x + (1/2) \times x^2 + (1/2 \cdot 3) \times x^3 + \dots\} + x^2 \{-1 - (3/4) \times x + \dots\}]$

3. $4(x^4 - x^2)y'' + x^3y' - y = 0$.

Ans. $y = ax^{1/2} \left(1 + \frac{1 \cdot 3}{4^2} x^2 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{4^2 \cdot 8^2} x^4 + \dots \right) + b \left[x^{1/2} \log x \left(1 + \frac{1 \cdot 3}{4^2} x^2 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{4^2 \cdot 8^2} x^4 + \dots \right) + 2x^{1/2} \left\{ \frac{1 \cdot 3}{4^2} \left(1 + \frac{1}{3} - \frac{1}{2} \right) x^2 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{4^2 \cdot 8^2} \left(1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} \right) x^4 + \dots \right\} \right]$

4. $xy'' + (p-x)y' - y = 0$ when (i) $p = 1$ (ii) p is not an integer.

Ans. (i) $y = a \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) + b \left[\log x \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) - \left\{ x + \frac{x^2}{1 \cdot 2} \left(1 + \frac{1}{2} \right) + \frac{x^3}{1 \cdot 2 \cdot 3} \left(1 + \frac{1}{2} + \frac{2}{3} \right) + \dots \right\} \right]$

(ii) $y = a \left(1 + \frac{x}{p} + \frac{x^2}{p(p+1)} + \dots \right) + b \left\{ \log x \left(1 + \frac{x}{p} + \frac{x^2}{p(p+1)} + \dots \right) + x^{1-p} \left(1 + x/1! + x^2/2! + \dots \right) \right\}$

5. $xy'' + y' + x^2y = 0$.

Ans. $y = au + bv$, where

$$u = 1 - \frac{x^3}{3^2} + \frac{x^6}{3^4(2!)^2} - \frac{x^9}{3^6(3!)^2} + \dots \quad \text{and} \quad v = u \log x + 2 \left[\frac{x^3}{3^3} - \frac{1}{3^5(2!)^2} \left(1 + \frac{1}{2} \right) x^6 + \dots \right]$$